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**Powered air purifying respirator for
particulate matter**

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Foreword

This Japanese Industrial Standard has been revised by the Minister of Health, Labour and Welfare and the Minister of Economy, Trade and Industry through deliberations at the Japanese Industrial Standards Committee as the result of proposal for revision of Japanese Industrial Standard submitted by Japan Safety Appliances Association (JSAA)/Japanese Standards Association (JSA) with the draft being attached, based on the provision of Article 12 Clause 1 of the Industrial Standardization Law applicable to the case of revision by the provision of Article 14. Consequently **JIS T 8157:2009** is replaced with this Standard.

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Powered air purifying respirator for particulate matter

1 Scope

This Japanese Industrial Standard specifies the powered air purifying respirators for particulate matter (hereafter referred to as PAPRs) used at the workplaces and other locations under the environment that may cause harmful effects on a human body due to the inhalation of particulate substances floating in air (hereafter referred to as the dust and the like).

This Standard is basically applicable to the conditions in which an electric fan of PAPR operates normally. The facepiece-type PAPR assumes the continuous use as a filtering device in the case where an electric fan is stopped due to the battery exhaustion and the wearer needs to move to the safe place, or an electric fan is intentionally stopped by the reason of the work.

WARNING 1 Do not use PAPR under the oxygen deficiency environment.

WARNING 2 Do not use PAPR in a place where a toxic gas or vapour, or volatile mist exists.

WARNING 3 Do not use PAPR under the environment in which an explosion may occur, unless it is proved as the explosion-proof electrical apparatus.

WARNING 4 Select PAPR correctly based on the type and concentration of the dust and the like.

Criteria for the selection of suitable respiratory protective devices are specified in **JIS T 8150**.

2 Normative references

The following standards contain provisions which, through reference in this text, constitute provisions of this Standard. The most recent editions of the standards (including amendments) indicated below shall be applied.

JIS C 1509-1 *Electroacoustics—Sound level meters—Part 1: Specifications*

JIS T 8001 *Glossary of terms for respiratory protective devices*

JIS T 8150 *Guidance for selection, use and maintenance of respiratory protective devices*

3 Terms and definitions

For the purpose of this Standard, the terms and definitions given in **JIS T 8001**, and the following apply.

3.1

minimum necessary air volume

minimum air volume specified in this Standard to be sent to the hood or a face shield of loose-fitting-type PAPR

3.2

nominal service life

period of time declared by the manufacturer as the service life of the battery-powered PAPR connected to the designated batteries, during which the facepiece-type PAPR can operate keeping its internal pressure of facepiece within the specified value or the loose-fitting-type PAPR can operate keeping its air volume within the specified value, when it is operated in an ordinary room (free from specially large amount of dust and the like)

3.3

respiratory inlet covering

generic term for facepieces, face shields and hoods mainly intended to be fitted onto the face for respiratory protection

NOTE In JIS T 8001, this term is defined as the generic term for facepieces, face shields, hoods and mouthpieces; however, PAPR does not have a mouth-piece.

3.4

fixture for air-flow resistance measurement

fixture attached to the facepiece-type PAPR used in the measurements of inhalation resistance, exhalation resistance, etc., having an air flow port and a pressure measuring port

4 Types

4.1 Types of PAPR

The types of PAPR shall be classified as follows based on the shape, performance of electric fan and leakage rate.

- a) **Classification by shape** Classified into the facepiece-type and the loose-fitting-type depending on the types of respiratory inlet covering to be used, and further divided into the front- or back-mounted type and the direct connection type. Table 1 shows the types of respiratory inlet covering used for respective shapes.

Table 1 Classification by shape

Shape		Type of respiratory inlet covering
Facepiece-type	Front- or back-mounted type	Full facepiece
		Half mask facepiece
	Direct connection type	Full facepiece
		Half mask facepiece
Loose-fitting-type	Front- or back-mounted type	Hood
		Face shield
	Direct connection type	Hood
		Face shield

- b) **Classification by performance of electric fan** Classified into the large air volume type and the normal air volume type according to the breathing condition in the tests.
- c) **Classification by leakage rate** Classified into Grade S, Grade A and Grade B depending on the leakage rate (see Table 2).

4.2 Types of particle filters

The particle filters used for PAPR shall be roughly classified into Type PL and Type PS according to the test particles, and further divided according to the filtering efficiency (see Table 3).

5 Performances

5.1 Leakage rate

The leakage rate of PAPR shall satisfy Table 2 when tested in accordance with 8.1.

Table 2 Leakage rate of PAPR

Grade by leakage rate	Leakage rate %
Grade S	0.1 max.
Grade A	1 max.
Grade B	5 max.

5.2 Filtering efficiency

The filtering efficiency of particle filters shall satisfy Table 3 when tested in accordance with 8.2.

Table 3 Filtering efficiency of particle filters

Type of particle filter		Filtering efficiency %	Test particles
Classification by test particles	Class according to filtering efficiency		
Type PL	PL3	99.97 min.	Dioctyl phthalate (DOP) particles
	PL2	99.0 min.	
	PL1	95.0 min.	
Type PS	PS3	99.97 min.	Sodium chloride (NaCl) particles
	PS2	99.0 min.	
	PS1	95.0 min.	

5.3 Internal pressure of facepiece of facepiece-type PAPR

The internal pressure of facepiece of facepiece-type PAPR shall satisfy Equation (1) when tested in accordance with 8.3.

If a manual flow rate controller is provided, the internal pressure shall satisfy Equation (1) for both of the minimum and maximum air volumes within the adjustable range of the controller.

$$0 \text{ (Pa)} < P_F < 400 \text{ (Pa)} \dots\dots\dots (1)$$

where, P_F : internal pressure of facepiece of facepiece-type PAPR (Pa)

5.4 Minimum necessary air volume for loose-fitting-type PAPR

The minimum necessary air volume for the loose-fitting-type PAPR shall satisfy Table 4 when tested in accordance with 8.4.

Table 4 Minimum necessary air volume for loose-fitting-type PAPR

Shape	Performance of electric fan	Minimum necessary air volume L/min
Loose-fitting-type	Large air volume type	138
	Normal air volume type	104

5.5 Noise level

The noise level of PAPR shall be 80 dB (A) or under when tested in accordance with 8.5.

5.6 Performance of facepiece-type PAPR under suspension of electric fan

5.6.1 Inhalation resistance

Under the condition where the electric fan is suspended, the inhalation resistance for the facepiece-type PAPR shall be 160 Pa or under when tested in accordance with 8.6.

5.6.2 Exhalation resistance

Under the condition where the electric fan is suspended, the exhalation resistance for the facepiece-type PAPR shall be 80 Pa or under when tested in accordance with 8.7.

5.6.3 Airtightness of exhalation valve

When the exhalation valve is tested in accordance with 8.8, the results shall satisfy the each of following:

- a) An internal pressure reduction starts immediately after the suction of air.
- b) The period is 15 s or longer from the stabilization of pressure reduction to the timing at which the pressure returns to ordinary pressure.

5.6.4 Elevated level of carbon dioxide concentration

Under the condition where the electric fan is suspended, the elevated level of carbon dioxide concentration shall be 2.0 % or under when tested in accordance with 8.9.

5.7 Strengths of head harness mount part and head harness

When tested in accordance with 8.10, a force required for the breakage or separation of the head harness mount part and the head harness shall satisfy Table 5.

Table 5 Strengths of head harness mount part and head harness

Type of facepiece	Force required for breakage or separation N
Full facepiece	50 min.
Half mask facepiece	25 min.

5.8 Strengths of breathing hose mount part and breathing hose

When tested in accordance with 8.11, a force required for the breakage or the separation of the breathing hose mount part and the breathing hose shall satisfy Table 6.

Table 6 Strengths of breathing hose mount part and breathing hose

Shape	Force required for breakage or separation N
Facepiece-type	100 min.
Loose-fitting-type	50 min.

6 Appearance and structures

6.1 Appearance

The appearance of PAPR shall be as follows.

- Free from cracks, smudges and abnormal warps.
- Each joint shall be properly connected.
- Components shall be properly installed.

6.2 General matters of structure

General matters of structure shall be as follows.

- No hindrance to the work of wearer under the worn condition.
- Capable of supplying the air to the wearer for breathing after the air in the workplace environment is purified by the particle filter.
- The joints shall be securely connected so as not to degrade the respiratory protection performance due to the leakage of air.
- The performance on usage shall not be interfered by possible impacts encountered during ordinary handling.
- Under a normal usage condition, the particle filter shall not fall off due to the impact, even if it contacts with an external structure.
- Giving no abnormal sense of oppression or pain when worn.

- g) A fuse or an alternative protective circuit shall be provided for the prevention of heating or ignition which may be caused by a short-circuit, etc.

6.3 Structures of PAPR classified according to shapes

The structures of PAPR classified according to shapes shall be as follows.

6.3.1 Facepiece-type PAPR

6.3.1.1 Facepiece-type PAPR of front- or back-mounted type

The facepiece-type PAPR of front- or back-mounted type shall consist of an electric fan, particle filter, breathing hose, facepiece, exhalation valve and head harness. It shall be so constructed that the purified air from which the dust and the like are removed by the particle filter is sent into the facepiece through the breathing hose by means of the electric fan, and at the same time, the exhaled air from the wearer and the excessive air are discharged to the outside air from the exhalation valve (see Figure 1 and Figure 2).

Even if the electric fan stops, the wearer can inhale the purified air by own inspiratory force.

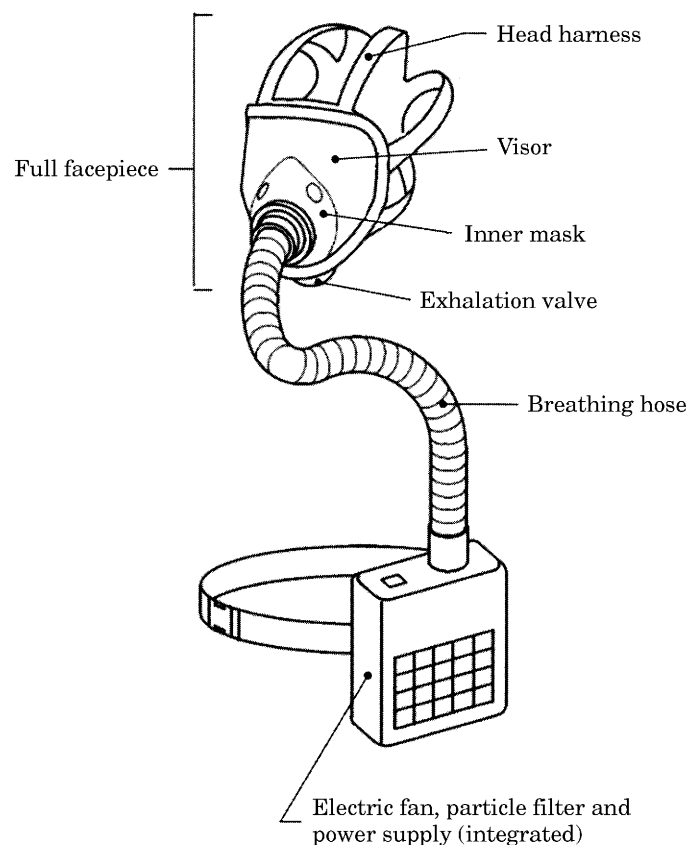


Figure 1 Example of facepiece-type PAPR of front- or back-mounted type having full facepiece

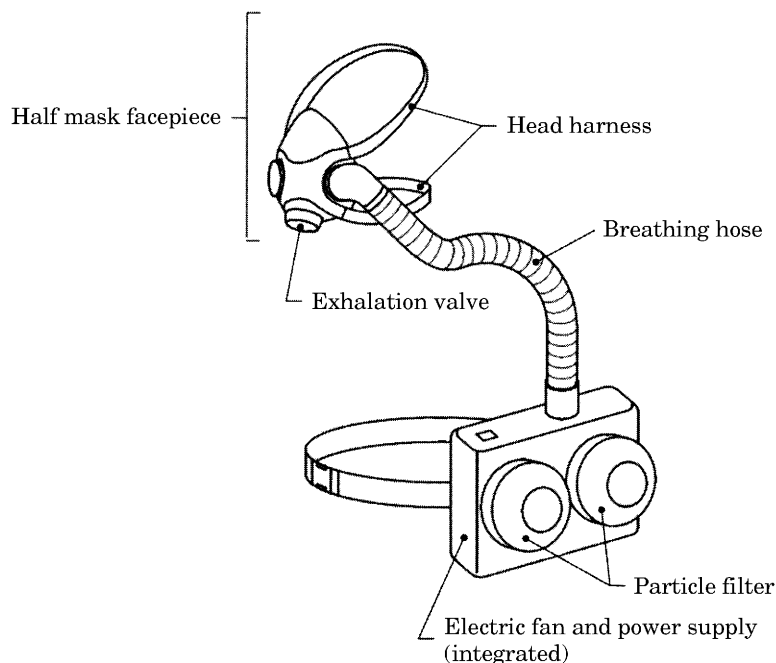


Figure 2 Example of facepiece-type PAPR of front- or back-mounted type having half mask facepiece

6.3.1.2 Facepiece-type PAPR of direct connection type

The facepiece-type PAPR of direct connection type shall consists of an electric fan, particle filter, facepiece, exhalation valve and head harness. It shall be so constructed that the purified air from which the dust and the like are removed by the particle filter is sent into the facepiece by means of the electric fan, and at the same time, the exhaled air from the wearer and the excessive air are discharged to the outside air from the exhalation valve (see Figure 3 and Figure 4). Even if the electric fan stops, the wearer can inhale the purified air by own inspiratory force.

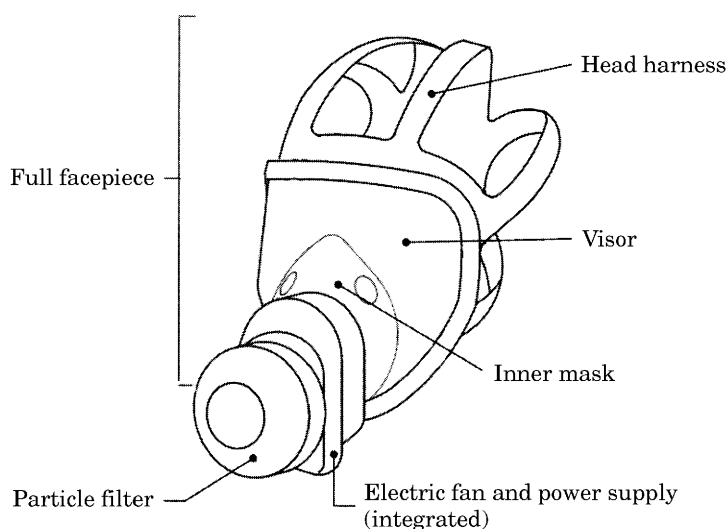


Figure 3 Example of facepiece-type PAPR of direct connection type having full facepiece

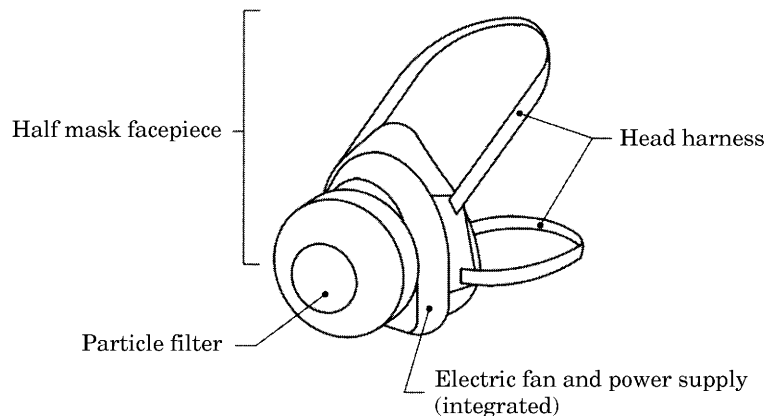


Figure 4 Example of facepiece-type PAPR of direct connection type having half mask facepiece

6.3.2 Loose-fitting-type PAPR

6.3.2.1 Loose-fitting-type PAPR of front- or back-mounted type

The loose-fitting-type PAPR of front- or back-mounted type shall consist of an electric fan, particle filter, breathing hose, and hood or face shield. It shall be so constructed that the purified air from which the dust and the like are removed by the particle filter is sent into the hood or the face shield through the breathing hose by means of the electric fan, and at the same time, the exhaled air from the wearer and the excessive air are discharged to the outside air from the clearance, etc. between the wearer's body and the hood or face shield (see Figure 5 and Figure 6). The following warning devices shall be provided according to the classification of leakage rate.

- Grade S and Grade A in leakage rate: alarm for air volume reduction
- Grade B in leakage rate: alarm for air volume reduction or alarm for power supply

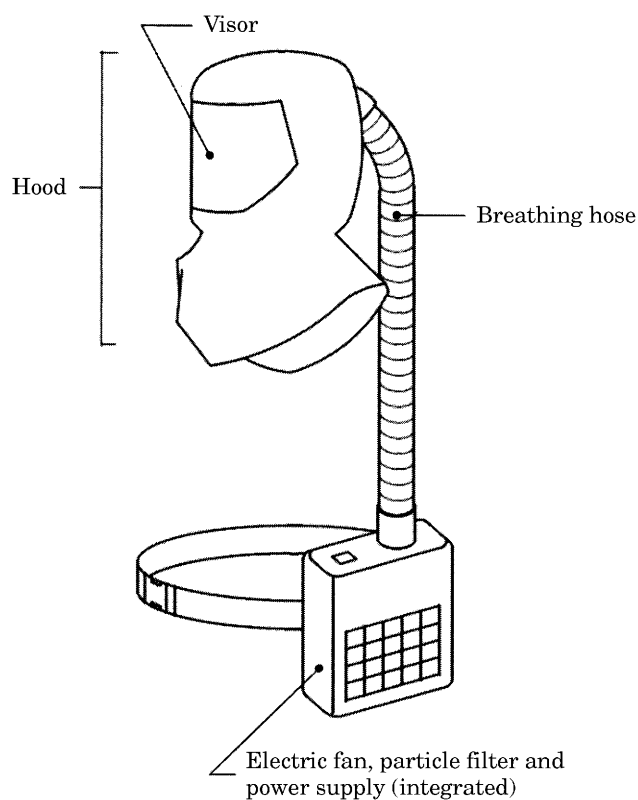


Figure 5 Example of loose-fitting-type PAPR having hood

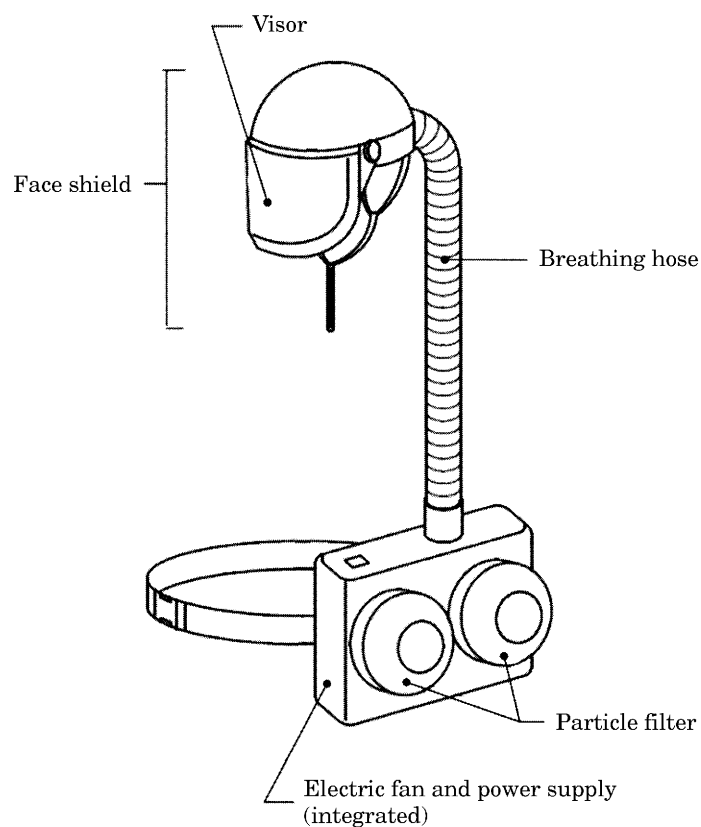


Figure 6 Example of loose-fitting-type PAPR having face shield

6.3.2.2 Loose-fitting-type PAPR of direct connection type

The loose-fitting-type PAPR of direct connection type shall consists of an electric fan, particle filter, and hood or face shield. It shall be so constructed that the purified air from which the dust and the like are removed by the particle filter is sent into the hood or the face shield by means of the electric fan, and at the same time, the exhaled air from the wearer and the excessive air are discharged to the outside air from the clearance, etc. between the wearer's body and the hood or face shield (see Figure 7). The following warning devices shall be provided according to the classification of leakage rate.

- Grade S and Grade A in leakage rate: alarm for air volume reduction
- Grade B in leakage rate: alarm for air volume reduction or alarm for power supply

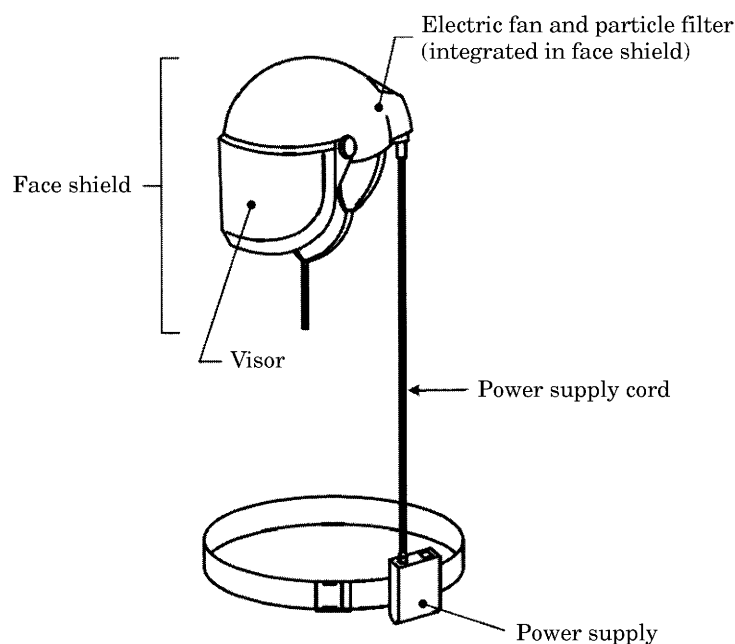


Figure 7 Loose-fitting-type PAPR of direct connection type having face shield

6.4 Structures of respective parts

6.4.1 Respiratory inlet covering

The structure of respiratory inlet covering shall be as follows.

- a) **Facepiece** Classified into two types of the full facepiece and the half mask facepiece. Both shall provide comfortable and close fit against the wearer's face. Also, the wearer can check its contact closeness against the face anytime by one-self. The visor for the full facepiece shall not be fogged easily due to the exhaled air.
- b) **Hood** To cover the head and the neck, or up to the body below shoulder, having such a structure as to prevent easy ingress of dust and the like. The visor shall not be fogged easily due to the exhaled air.

- c) **Face shield** To cover the entire face of the wearer, the circumference of which is surrounded with a plastic or rubber cheek pad to have such a structure as to prevent easy ingress of dust and the like into the face shield through its clearance. The visor shall not be fogged easily due to the exhaled air.

6.4.2 Electric fan

The structure of electric fan shall be as follows.

- a) No hindrance due to water, dust and the like for operation.
- b) Provided with a means of verifying its air flow volume before use. However, this requirement is not necessarily be applicable to PAPR having the following measures or alarms.
- Facepiece-type PAPR having a check method that the internal pressure of facepiece is positive pressure when worn
 - Facepiece type PAPR having alarm for internal pressure reduction
 - Loose-fitting-type PAPR having alarm for air volume reduction

6.4.3 Exhalation valve

The structure of exhalation valve shall be as follows.

- a) Shall operate reliably and accurately as the normal respiration, irrespective of the humidity of the valve and the valve seat, under the condition where the electric fan is suspended.
- b) Capable of keeping the closed condition, irrespective of the facepiece direction, when the inner and outer pressures are balanced.
- c) Having a protection by a cover, etc. to prevent the damage due to the external force.

6.4.4 Breathing hose

The structure of breathing hose shall be as follows.

- a) Not interference with the wearer's movement, and no hindrance to the air flow even if it is bent into various states, when worn.
- b) No hindrance to the air flow even if it is oppressed by wearer's chin, arm or the like.
- c) Having a length not hindering the free movement of neck.

6.4.5 Power supply

The structure of power supply shall be as follows.

- a) The power supply shall be batteries or an external power supply. The voltage near the body of wearer shall be 24 V or under.
- b) Its cord shall be hard to be cut, even if it is caught by a protrusion or the like during work.
- c) When batteries are used, they do not fall off due to an impact or the like during work.

- d) The connector shall be so constructed that it does not come off due to an impact or the like during work.
- e) The switch shall not be easily turned off even if the power supply touches something in the workplace environment during its use.

6.4.6 Particle filter

The structure of particle filter shall be as follows.

- a) Capable of removing the flow-in dust and the like from the environmental air.
- b) To be easily and reliably replaceable.

6.4.7 Alarm

The alarm shall be as follows.

- a) **Alarm for internal pressure reduction** Used for the facepiece-type PAPR, having a function to alert the wearer during the use that the internal pressure of facepiece is approaching or has reached the lower limit specified in **5.3**.
- b) **Alarm for air volume reduction** Used for the loose-fitting-type PAPR, having a function to alert the wearer during the use that the air flow volume is approaching the minimum necessary air volume specified in **5.4**.
- c) **Alarm for power supply** Having a function to alert the wearer during the use that the battery voltage has reached the lower voltage limit above which PAPR works effectively.

6.4.8 Charger

When the PAPR whose power supply is a rechargeable battery, the charger shall be as follows.

- a) The batteries cannot be set in the charger with their polarities reversed.
- b) The charger shall be provided with a safety device against the overcharge or the overcurrent for the prevention of abnormal heat generation, unless the battery is equipped with a safety device.

6.4.9 Head harness

The head harness shall have an adequate elasticity and strength, and the length shall be adjustable.

7 Materials

The materials used for respective parts of PAPR shall be as follows.

- a) Having adequate strength and elasticity, etc. for the applications.
- b) The materials used for the areas in contact with the skin shall not have harmful influence on the skin, and be capable of sterilizing.
- c) The materials used in the air-flow passage shall not generate any hazardous substances.

- d) The exposed metal part shall be corrosion-resistant or subjected to an anticorrosion treatment.

8 Tests

8.1 Leakage rate test

8.1.1 Overview

PAPR shall be activated in a chamber in which test particles are generated, mounted on a swingable head and torso simulator connected to a breathing machine, then the leakage rate shall be calculated from the test particle concentrations in the chamber and the inhalation passage.

8.1.2 Test conditions

8.1.2.1 Power supply

The power supply designated for PAPR shall be used for the test. If the rechargeable batteries are to be used, they shall be fully charged.

8.1.2.2 Particle filter

An unused product of particle filter designated for PAPR shall be used.

8.1.2.3 Conditions of test particles

8.1.2.3.1 Type

The test particles to be used shall be sodium chloride particles.

8.1.2.3.2 Production method and concentration

The production method and concentration of NaCl particles shall be as follows.

- a) NaCl particles shall be produced by spraying NaCl aqueous solution, then being dried.
- b) A median in number-particle size distribution shall be $0.06\ \mu\text{m}$ to $0.2\ \mu\text{m}$, and the geometrical standard deviation shall be 2.0 or under.
- c) The sodium chloride concentration in the chamber shall be $12\ \text{mg}/\text{m}^3 \pm 6\ \text{mg}/\text{m}^3$ at the location close to the front surface of respiratory inlet covering without the influence of the exhaust stream from PAPR. The variation shall fall within $\pm 15\ \%$.

8.1.2.4 Test particle concentration meter

The particle concentration meter shall be of light scattering type or that having at least equivalent performance thereto, and the lower limit of determination shall be $6 \times 10^{-4}\ \text{mg}/\text{m}^3$ or under.

8.1.2.5 Operating conditions of breathing machine

The operating conditions of breathing machine shall be the performance of electric fan marked on PAPR as shown in Table 7.

Table 7 Operating conditions of breathing machine

Operating condition of breathing machine	Performance of electric fan	
	Large air volume type	Normal air volume type
Respiratory waveform	Sinusoidal wave	
Ventilation amount per breath L/number of times	$1.6 \pm (5 \%)$	$1.5 \pm (5 \%)$
Number of breaths per minute Number of times/min	25 ± 1	20 ± 1

8.1.2.6 Swingable head and torso simulator

The swingable head and torso simulator used in the tests shall be as follows.

- a) **Configuration** Figure 8 shows the configuration of swingable head and torso simulator. Such a swingable head and torso simulator is applicable that an inhalation port and an exhalation port are attached to the mouth of dummy head specified in Figure 9.

Unit: mm

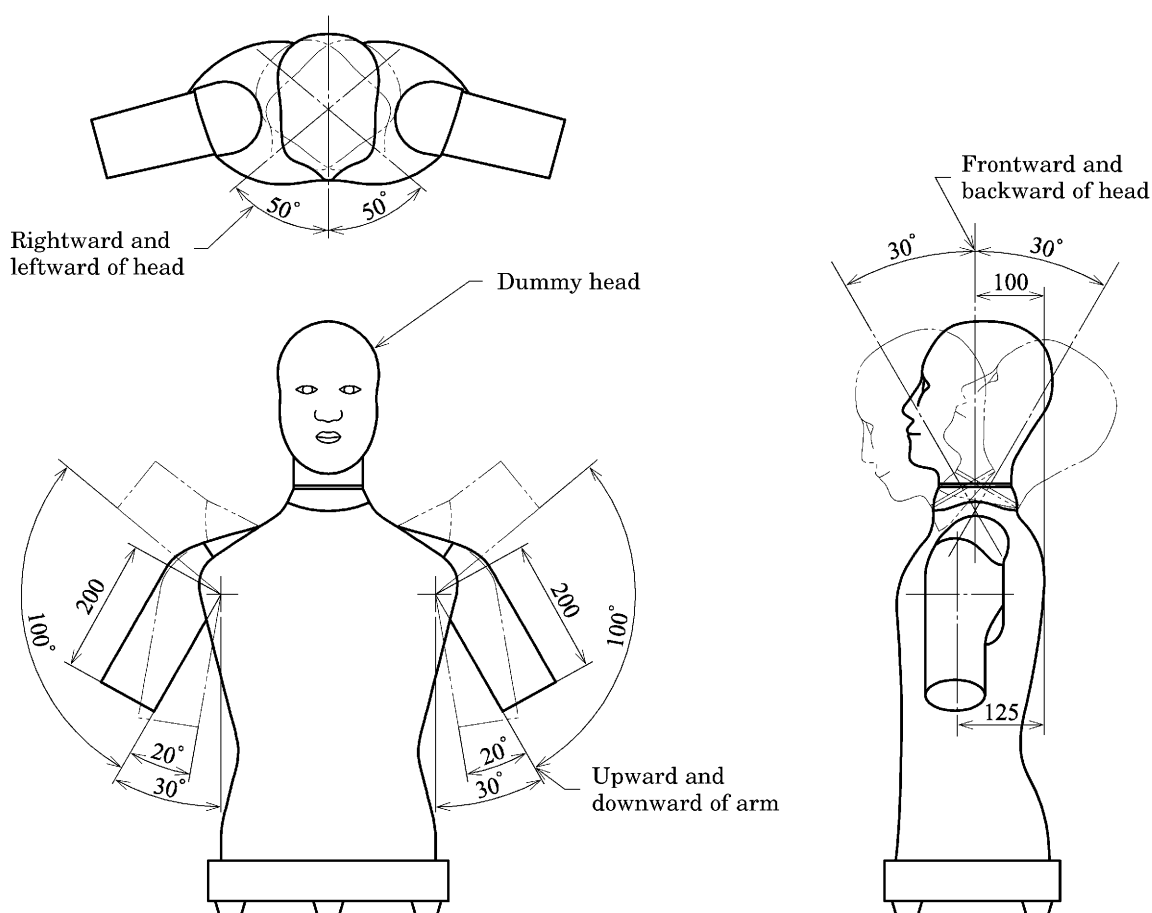


Figure 8 Swingable head and torso simulator

Unit: mm

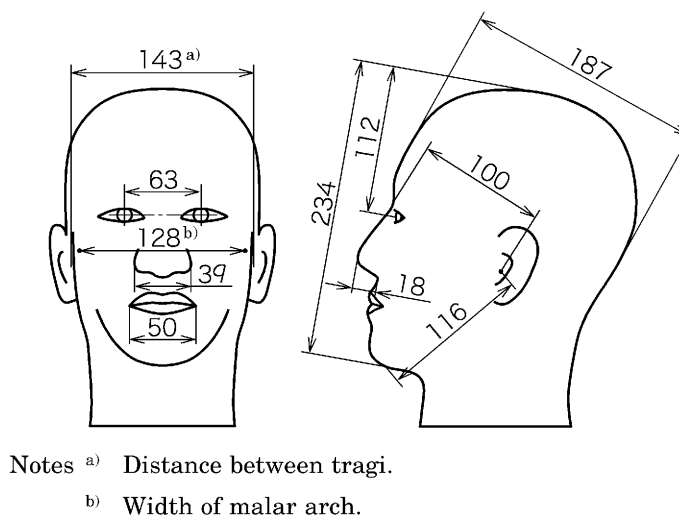


Figure 9 Dummy head

- b) **Conditions of swingable parts** Table 8 shows the swingable points, directions and ranges, and periods.

Table 8 Conditions of swingable parts

Swingable mode	Swingable points	Swingable direction	Swingable range	Period Number of times/min
Head-forward and backward	Head	Forward and backward	Forward and backward 30° with respect to vertical upward, respectively	17 ± 1
Head-rightward and leftward		Rightward and leftward rotation	Rightward and leftward 50° with respect to horizontal forward, respectively	11 ± 1
Upper arm-upward and downward	Arm	Upward and downward in rightward and leftward	Between 10° and 130° with respect to vertical downward	7 ± 1

8.1.3 Test procedure

The test procedure shall be as follows (see Figure 10).

- Attach the particle filter to PAPR.
- Connect the power supply to PAPR.
- Mount PAPR on the swingable head and torso simulator connected to the breathing machine. At this time, do not use a sealant or the like between the respiratory inlet covering and the swingable head and torso simulator.
- Blow the air toward the respiratory inlet from the lower front position 1 m or more apart from the respiratory inlet in an angle of 45°. Adjust the air velocity to 0.5 m/s or more at the location 30 cm windward.

- e) Adjust the test particle concentration to the value specified in **8.1.2.3.2 c)** at the location close to the front surface of respiratory inlet covering without the influence of the exhaust stream from PAPR (hereafter referred to as test particle concentration in chamber).
- f) Turn on the electric fan of PAPR. If PAPR has a manual flow rate controller, set the flow rate to the minimum within the range specific to the controller.
- g) Activate the breathing machine and the swingable head and torso simulator (select one swingable mode specified in Table 8).
- h) Taking the timing 3 min after the activation of the head and torso simulator as the measurement start, measure the NaCl particle concentration in the chamber (*WO*) and the NaCl particle concentration in the inhalation passage (*WI*) for 1 min. Furthermore, measure *WO* and *WI* continuously for 1 min. In these measurements, the unit for NaCl particle concentration shall be mg/m^3 .

In the case where such a test particle concentration meter of which one time measuring period is shorter than 1 min is used, determine the number (*N*) by truncating the quotient to an integer, which is obtained by dividing 1 min by the one-time measuring period. From the measurement start, measure the NaCl particle concentrations in the chamber and in the inhalation passage *N* times in succession. Take each integrated value of the measured values as *WO* and *WI* from the starting point of the measurement. Similarly, in the same mode, measure the NaCl particle concentrations *N* times in succession from 1 min after the measurement start, and each integrated value shall be taken as *WO* and *WI* from 1 min after the starting point of the measurement.

- i) Stop the operation of the breathing machine and the swingable head and torso simulator, and turn off the electric fan of PAPR.
- j) For each of two other swingable modes in Table 8, carry out the same procedure, respectively.

When changing the swingable mode, the following matters should be considered.

- Replace the particle filter to new when the swingable mode is changed.
- PAPR need not be restored as far as the factor for test condition does not change, even if the swingable mode is changed. However, in such a case where PAPR position needs to be changed for the filter replacement or PAPR is unintentionally touched by the person in charge of testing, restore PAPR or take another measures.
- k) Substitute the series of values into WO_x and WI_x ($x=1$ to 6) in Equation (2), which are numbered from 1 to 6 in the measured order obtained as *WO* and *WI* (six for each), and obtain the leakage rate [L_x % ($x=1$ to 6)] on each measurement. Take the largest among a series of the calculated values (6 in total) as the leakage rate of the said test.

$$L_x = \frac{WI_x}{WO_x} \times 100 \dots\dots\dots (2)$$

where, L_x : leakage rate of xth measurement (%)

WI_x : test particle concentration of xth measurement in
 inhalation passage (mg/m^3)

WO_x : test particle concentration of xth measurement in
 chamber (mg/m^3)

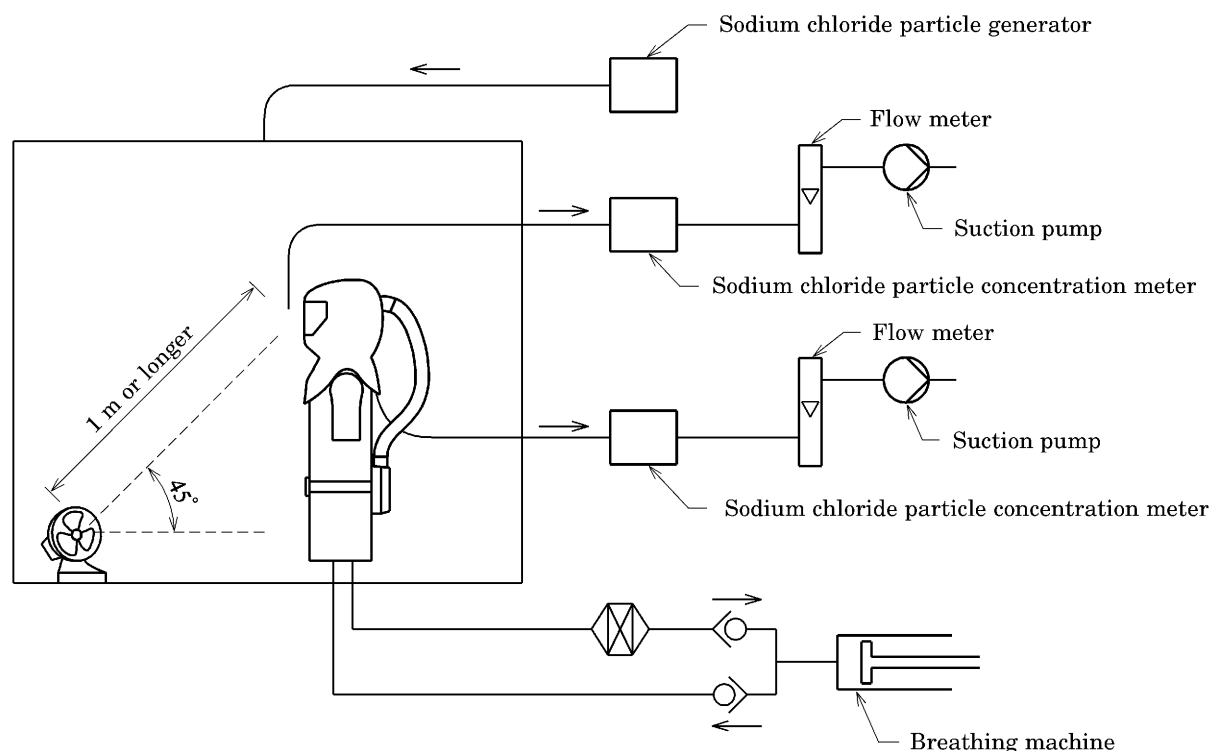


Figure 10 Example of test apparatus for leakage rate

8.2 Filtering efficiency test

8.2.1 Overview

The particle filter to be attached to PAPR is used as a specimen, the filtering efficiency is measured in succession at a constant period while a steady air flow containing test particles is supplied until the cumulative supplied quantity of test particles achieves a certain amount, and the minimum filtering efficiency during the test shall be determined.

8.2.2 Test conditions

8.2.2.1 Specimen

An unused particle filter designated for PAPR shall be used as the specimen.

8.2.2.2 Test apparatus

The test apparatus shall consist of a test particle generator, particle concentration meters, a chamber for specimen, a specimen holder, a flow meter, a suction pump, and so on (see Figure 11). The particle concentration meter that employs light scattering method shall be used. The air flow port of specimen holder and the outflow port of the chamber for specimen shall be so constructed that the airtightness of the joints can be kept when they are connected.

8.2.2.3 Test particles

The test particles shall be dioctyl phthalate (DOP) particles or sodium chloride (NaCl) particles specified in Table 3 depending on classes marked on the specimen. Each of the test particles shall comply with the following requirements.

a) **DOP particles** Shall be as follows.

- 1) Particles shall be produced by spraying DOP.
- 2) A median in number-particle size distribution shall be $0.15\ \mu\text{m}$ to $0.25\ \mu\text{m}$, and the geometrical standard deviation shall be 1.6 or under.
- 3) The particle concentration of DOP in the chamber shall be $100\ \text{mg}/\text{m}^3$ or under, and the variation shall fall within $\pm 15\ \%$.

b) **NaCl particles** Shall be as follows.

- 1) Particles shall be produced by spraying NaCl aqueous solution, then being dried.
- 2) A median in number-particle size distribution shall be $0.06\ \mu\text{m}$ to $0.1\ \mu\text{m}$, and the geometrical standard deviation shall be 1.8 or under.
- 3) The particle concentration of NaCl in the chamber shall be $50\ \text{mg}/\text{m}^3$ or under, and the variation shall fall within $\pm 15\ \%$.

8.2.2.4 Test flow rate

The test flow rate shall be as given in Table 9. In the case where two or more particle filters are attached to PAPR, any of the following shall apply.

- a) The particle filters to be attached to PAPR, as a whole, shall satisfy the relevant flow rate shown in Table 9.
- b) The flow rate for a single particle filter shall be obtained by dividing the relevant flow rate in Table 9 by the number of filters attached.

Table 9 Flow rate of particle filter in filtering efficiency test

Performance of electric fan	Test flow rate L/min
Large air volume type	138
Normal air volume type	104

8.2.2.5 Supplied quantity of test particles

The filtering efficiency test on filters shall be continued at least until the supplied quantity of particles reaches the value specified in Table 10. When the test flow rate is in accordance with **8.2.2.4 a)**, the value shall be the relevant supplied quantity of test particles specified in Table 10. When in accordance with **8.2.2.4 b)**, the value shall be the supplied quantity of test particles which is obtained by dividing the relevant supplied quantity of test particles specified in Table 10 by the number of chemical cartridge with filters attached to PAPR.

Table 10 Supplied quantity of test particles

Type of particle filter	Test particles	Supplied quantity of test particles mg
Type PL	DOP particles	400
Type PS	NaCl particles	200

8.2.3 Test procedure

The test procedure shall be as follows (see Figure 11).

- Attach the specimen on the specimen holder, then, place it in the chamber for specimen. Keep the airtightness between the specimen and the specimen holder using a sealant or the like.
- Pass the air containing test particles through the specimen at the flow rate specified in 8.2.2.4.
- One minutes after the start of air flow, continuously measure the test particle concentration on the upstream and downstream sides thereof with the particle concentration meter at every 5 min or under until the supplied quantity of test particles has reached the value specified in Table 10. Calculate the filtering efficiency using Equation (3) on a period-by-period basis to obtain the minimum value of the filtering efficiency.

$$E = \frac{C_o - C_i}{C_o} \times 100 \dots\dots\dots (3)$$

where, E : filtering efficiency in each period (%)
 C_o : test particle concentration before passing through specimen in each period (mg/m^3)
 C_i : test particle concentration after passing through the specimen in each period (mg/m^3)

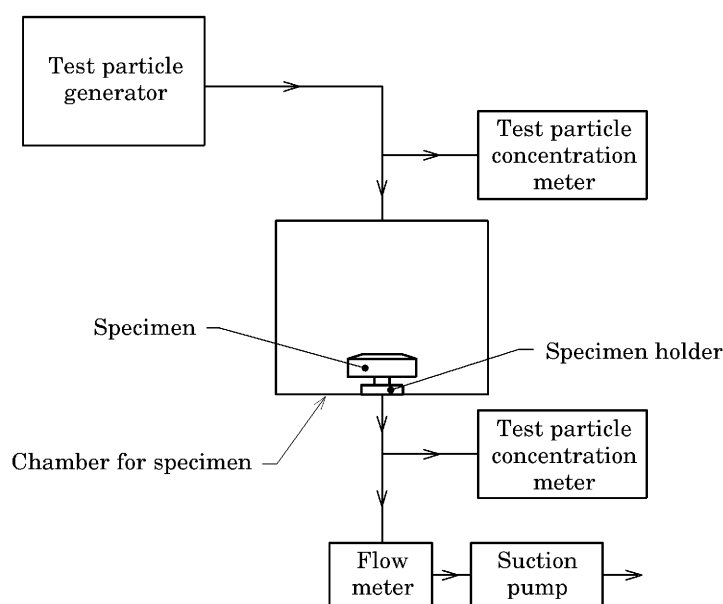


Figure 11 Example of test apparatus for filtering efficiency

8.3 Internal pressure test on facepiece

8.3.1 Test conditions

8.3.1.1 Specimen

An facepiece-type PAPR is used as the specimen.

8.3.1.2 Power supply

The power supply designated for PAPR shall be used for the test. If the rechargeable batteries are to be used, they shall be fully charged.

8.3.1.3 Particle filter

An unused product of particle filter designated for PAPR shall be used.

8.3.1.4 Test apparatus

The test apparatus consists of a dummy head, breathing machine, fine pressure differential gauge, recorder or data processor, and so on (see Figure 12). The dummy head specified in Figure 9 shall be used. An air flow port shall be attached to the mouth and a pressure measuring port, to the nostril part. A fine pressure differential gauge to be used shall be capable of measuring the target range with a sensitivity of 1 Pa. The whole time constant of the fine pressure differential gauge including recorder or data processor shall be 0.2 s or under.

8.3.2 Test procedure

The test procedure shall be as follows.

- a) Put the facepiece on the dummy head as shown in Figure 12. At this time, keep the airtightness between the face seal and the dummy head using a sealant or the like.
- b) Activate the breathing machine and PAPR. The operating conditions of the breathing machine shall be as given in Table 7 depending on the performance of electric fan designated for PAPR. If PAPR has a manual flow rate controller, carry out c) by setting the flow rate to the maximum and the minimum within the range specific to the controller.
- c) Activate PAPR (after the flow rate is changed if manual flow rate controller is provided), wait until the breathing machine takes about 6 breaths and measure the internal pressure of facepiece continuously for 1 min with the fine pressure differential gauge.

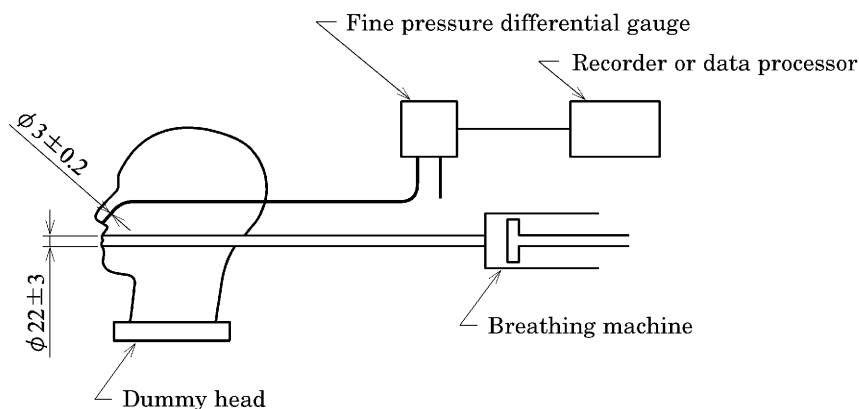


Figure 12 Example of test apparatus for internal pressure of facepiece-type PAPR

8.4 Minimum necessary air volume for loose-fitting-type PAPR

8.4.1 Test conditions

8.4.1.1 Specimen

An loose-fitting-type PAPR is used as the specimen.

8.4.1.2 Power supply

The power supply designated for PAPR shall be used for the test. If the rechargeable batteries are to be used, they shall be fully charged.

8.4.1.3 Particle filter

An unused product of particle filter designated for PAPR shall be used.

8.4.2 Test apparatus

The test apparatus consists of an airtight chamber, head and torso simulator (or dummy head), flow meter, suction device, fine pressure differential gauge, and so on (see Figure 13). On the side face of the airtight chamber, depending on the specimen's shape, a hole for piercing the breathing hose or a hole for attaching the inhalation port of PAPR from inside of the chamber shall be provided. The usage conditions of the head and torso simulator and the dummy head shall be as follows.

- **Head and torso simulator** Capable of mounting all specimens
- **Dummy head** Using only when the area covered by hood or face shield of specimen is in the range from the head top to the neck part.

8.4.3 Test procedure

The test procedure shall be as follows.

- a) Mount the hood or the face shield on the head and torso simulator or the dummy head.
- b) Install the specimen as follows depending on whether it is a front- or back-mounted type or a direct connection type.

- **Front- or back-mounted type** With the electric fan or the like which is connected to the breathing hose being outside of the airtight chamber, connect the other end of the breathing hose to the hood or the face shield. Keep the airtightness of clearance between the breathing hose and the airtight chamber using a sealant or the like.
 - **Direct connection type** Connect the inhalation port of PAPR to the hole of the airtight chamber from inside, directly or through a dedicated tube. Keep the airtightness of clearance between the breathing hose and the airtight chamber using a sealant or the like.
- c) Activate PAPR. If PAPR has a manual flow rate controller, set the flow rate to the minimum within the range specific to the controller.
- d) Activate the suction device.
- e) Adjust the air flow volume so that the fine pressure differential gauge gives 0 Pa. Read the indicated value on the flowmeter after 5 min from PAPR activation, and take this value as the air volume for PAPR.

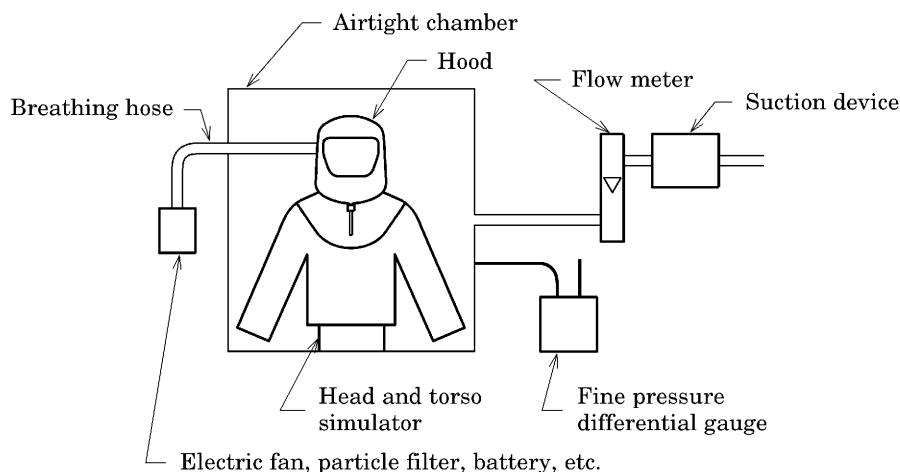


Figure 13 Example of test apparatus for minimum necessary air volume for loose-fitting-type PAPR

8.5 Noise level measurement

8.5.1 Measuring conditions

8.5.1.1 Power supply

The power supply designated for PAPR shall be used for the test. If the rechargeable batteries are to be used, they shall be fully charged.

8.5.1.2 Particle filter

An unused product of particle filter designated for PAPR shall be used.

8.5.1.3 Measuring device

The measuring device to be used shall be as follows.

- a) **Breathing machine** Activated under the conditions specified in Table 7.
- b) **Dummy head** Worked such that an air flow port is attachable to the mouth of dummy head specified in Figure 9, and the sensor of sound level meter is attachable to the ears (hereafter referred to as dummy head for noise).
- c) **Noise meter** That of Class 1 or Class 2 specified in **JIS C 1509-1**, capable of measuring sound levels with a constant interval not exceeding 100 μ s, and having any of the following functions.
 - Capable of indicating a maximum value during 1 min
 - Capable of recording the measured value with a constant interval not exceeding 100 ms.

8.5.2 Test procedure

The test procedure shall be as follows.

- a) Connect the air flow port of dummy head for noise with the breathing machine, and attach the noise sensor of sound level meter to the ears of dummy head for noise, or confirm that the sensor is being in the position.
- b) Mount the specimen on the dummy head for noise.
- c) Activate the breathing machine. The operating conditions of the breathing machine shall be as given in Table 7 depending on the performance of electric fan designated for PAPR.
- d) Activate PAPR. If PAPR has a manual flow rate controller, set the flow rate to the maximum within the range specific to the controller.
- e) After 5 min from the activation of PAPR, measure the noise level for 1 min at left and right ear parts of dummy head for noise at a constant cycle not exceeding 100 μ s. The measurement may be carried out at left ear part and right ear part separately using one measuring device, or simultaneously at both parts by using two measuring devices.
- f) Determine the maximum among these measured values (for the sound level meter capable of recording measured data at a constant interval not exceeding 100 ms, the maximum peak value among the cyclic noise levels during 1 min), and take it as the noise level of PAPR. If an external sudden noise has possibly affected the measured values, discard the measured data, and measure in a same manner one more time.

8.6 Inhalation resistance test

8.6.1 Specimen

A whole facepiece-type PAPR is used as the specimen under the condition where the electric fan is suspended.

8.6.2 Test apparatus

The test apparatus shall consist of a fixture for air-flow resistance measurement, a suction pump, a flow meter, a fine pressure differential gauge, and so on (see Figure 14).

8.6.3 Test procedure

8.6.3.1 Case where facepiece-type PAPR having full facepiece without inner mask or having half mask facepiece is used as specimen

The test procedure shall be as follows.

- a) Mount the specimen on the fixture for air-flow resistance measurement, and seal the contact part to the fixture for air-flow resistance measurement with putty or the like to prevent any leakage.
- b) Activate the suction pump, suction the specimen inside with the flow rate 40 L/min, and measure the pressure difference between inside and outside of the specimen.

8.6.3.2 Case where facepiece-type PAPR having full facepiece with inner mask is used as specimen

The test procedure shall be as follows.

- a) Mount the specimen on the fixture for air-flow resistance measurement, and examine for any clearance between the inner mask and the fixture for air-flow resistance measurement.
- b) If there is no clearance between the inner mask and the fixture for air-flow resistance measurement, carry out the same measurement procedure as that in **8.6.3.1**.
- c) If there is a clearance between the inner mask and the fixture for air-flow resistance measurement, remove the inner mask from the specimen, then, carry out the test following the same test procedure as that in **8.6.3.1** on each of the specimen from which the inner mask is removed and the inner mask, and take the sum of these measurement results as the inhalation resistance.

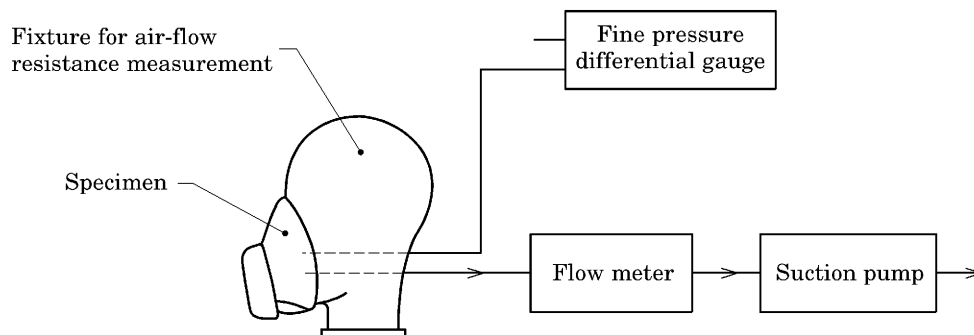


Figure 14 Example of inhalation resistance tester

8.7 Exhalation resistance test

8.7.1 Specimen

A whole facepiece-type PAPR is used as the specimen under the condition where the electric fan is suspended.

8.7.2 Test apparatus

The test apparatus shall consist of a fixture for air-flow resistance measurement, exhalation pump, flow meter, a fine pressure differential gauge, and so on (see Figure 15).

8.7.3 Test procedure

The test procedure shall be as follows.

- Mount the specimen on the fixture for air-flow resistance measurement, and seal the contact part to the fixture for air-flow resistance measurement with putty or the like to prevent any leakage.
- Activate the exhalation pump, and flow the air toward the specimen with the flow rate 40 L/min.
- Measure the pressure difference between inside and outside of the specimen with the fine pressure differential gauge.

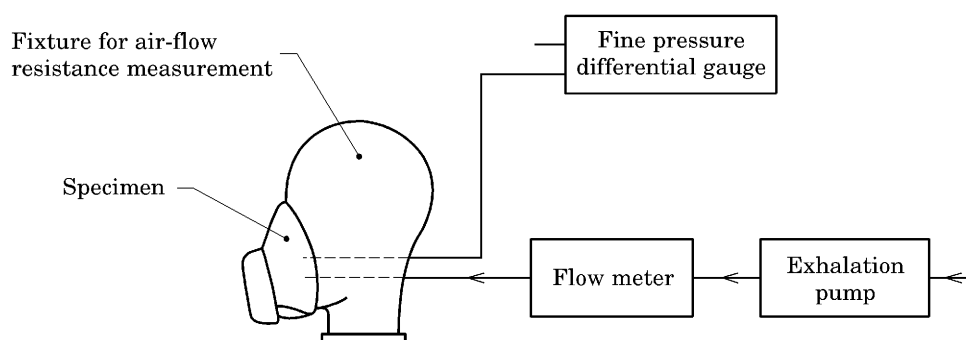


Figure 15 Example of exhalation resistance tester

8.8 Airtightness test of exhalation valve

8.8.1 Specimen

The specimen shall be the exhalation valve and the exhalation valve seat of facepiece-type PAPR under test.

8.8.2 Test apparatus

The test apparatus shall consist of an airtightness testing device, manometer, switching valve, flow meter and suction pump (see Figure 16). The inner volume of an airtightness testing device shall be $50 \text{ cm}^3 \pm 5 \text{ cm}^3$.

8.8.3 Test procedure

The test procedure shall be as follows.

- Mount the specimen on the airtightness testing device.
- Open the switching valve.
- Activate the suction pump, and suck the air from the airtightness testing device with the flow rate 1 L/min. Examine the reduced internal pressure of the airtightness testing device by closing the specimen.

- d) Continue the suction further to make the pressure inside the airtightness testing device 1 470 Pa lower than the outside pressure, and close the switching valve. Leave the airtightness testing device as it is, and measure the period of time required for the internal pressure of airtightness testing device to return to the ordinary pressure.

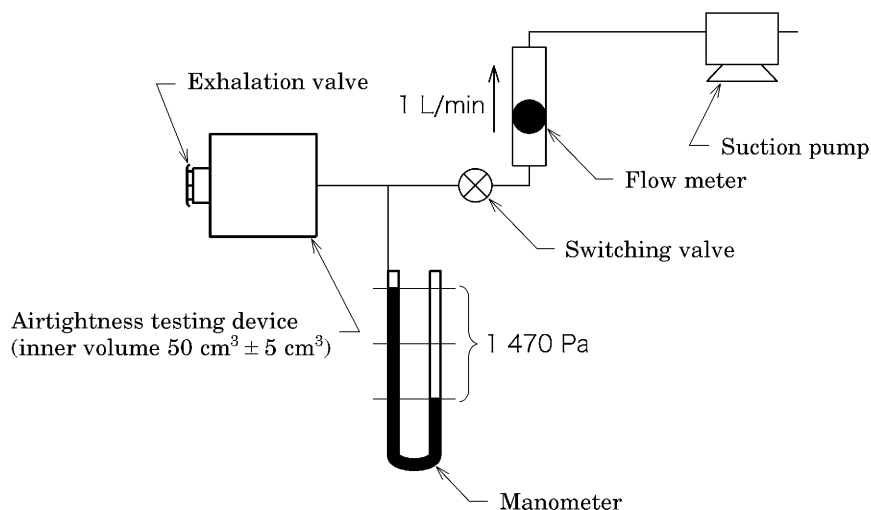


Figure 16 Example of airtightness test apparatus for exhalation valve

8.9 Elevated level test of carbon dioxide concentration

8.9.1 Overview

Simulate breaths using the breathing machine through a dummy head. Use the air containing carbon dioxide 5 % in concentration as the expiration. Use the testing device that measures the carbon dioxide concentration in the inhalation when the exhalation above is supplied under both the conditions where the facepiece-type PAPR is mounted on the dummy head and that not mounted, then obtain the elevated level of carbon dioxide concentration from the difference between them.

8.9.2 Specimen

A whole facepiece-type PAPR is used as the specimen under the condition where the electric fan is suspended.

8.9.3 Test apparatus

The breathing passage shall be so constructed that a dummy head specified in Figure 9 with an inhalation port and an exhalation port fitted to the mouth part are connected to the breathing machine through a check valve for each, and an inhalation gas collection bag and an expiration gas feeding bag are connected to the inhalation passage side and the exhalation passage side, respectively (see Figure 17). A blower shall be installed in front of the dummy head. A carbon dioxide concentration measuring device shall be used to measure the carbon dioxide concentration in the inhalation gas collection bag, which is capable of measuring the concentration in the range 0.1 % to 5 %.

8.9.4 Test procedure

The test apparatus for the elevated level of carbon dioxide concentration in inhalation air shown in Figure 17 shall be used. The test procedure shall be as follows.

- a) Adjust the blower so that the wind velocity at the mouth part of dummy head is approximately 0.5 m/s.
- b) Fill the expiration gas feeding bag with the air containing carbon dioxide 5 % in concentration.
- c) Empty the inhalation gas collection bag.
- d) Activate the breathing machine at the rate (2.0 ± 0.1) L/cycles $\times$ (15 ± 1) cycles/min under the condition without the specimen on the dummy head.
- e) Measure the carbon dioxide concentration in the inhalation gas collection bag, and take the value as “the carbon dioxide concentration in the inhalation air under the condition where the specimen is mounted” (C_R).
- f) Subsequently, operate the breathing machine at the rate (2.0 ± 0.1) L/cycles $\times$ (15 ± 1) cycles/min under the condition with the specimen on the dummy head.
- g) Measure the carbon dioxide concentration in the inhalation gas collection bag, and take this value as “the carbon dioxide concentration in the inhalation air under the condition where the specimen is not mounted” (C_N).
- h) Substitute the values obtained in **e**) and **g**) into C_N and C_R , respectively, in Equation (4) to determine the elevated level of carbon dioxide in the inhalation air.

$$D = C_R - C_N \dots\dots\dots (4)$$

where, D : elevated level of carbon dioxide concentration in inhalation by PAPR (%)

C_R : carbon dioxide concentration in inhalation with specimen mounted on dummy head (%)

C_N : carbon dioxide concentration in inhalation with specimen not mounted on dummy head (%)

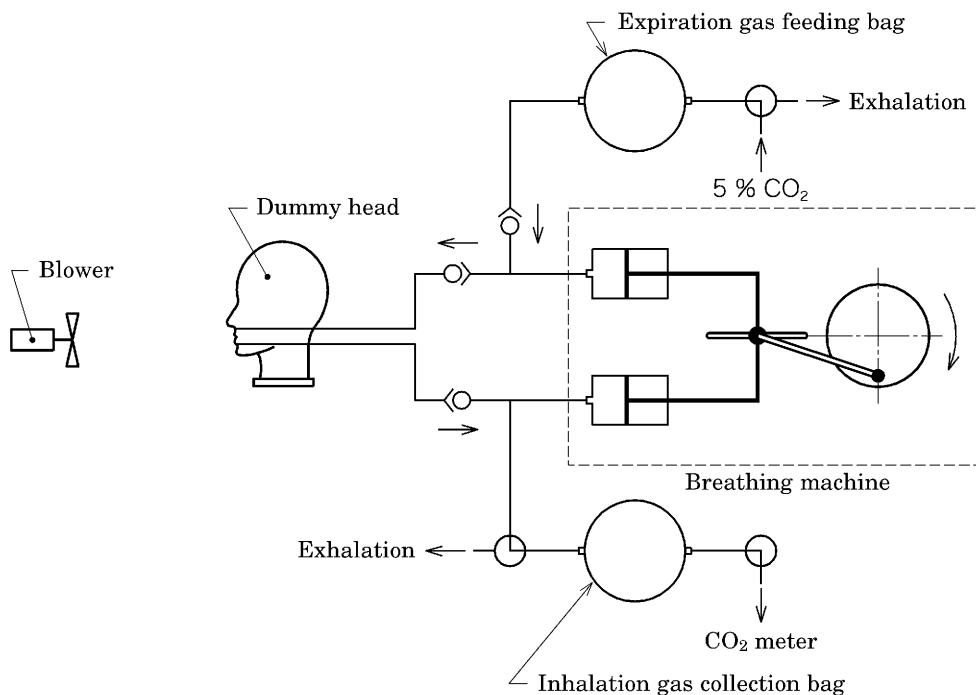


Figure 17 Example of testing apparatus for elevated level of carbon dioxide concentration

8.10 Strength test for head harness mount part and head harness

The strength test for a head harness mount part and a head harness shall be in accordance with either of **a)** and **b)** below.

- a) **Method applying constant force** Apply the tensile force for 10 s between the facepiece (a part necessary for fixing is permissible) and the head harness end (one position of other mount side) 50 N for the full facepiece, and 25 N for the half mask facepiece to every head harness and its mount part. Then, examine for any breakage or separation.
- b) **Method using tensile tester** Attach the facepiece (a part necessary for fixing is permissible) and the head harness end (one position of other mount side) to the tensile tester, pull them at a speed of 20 cm/min, and measure the force when the head harness or its mount part breaks or separates, for every head harness and its mount part. If the force applied by the tensile tester exceeds 100 N for the full facepiece and 50 N for half mask facepiece, and if no breakage or separation seems to occur easily with additional force application, the test may be interrupted by recording the final test force applied and the fact that no breakage or separation has occurred.

8.11 Strength test for breathing hose mount part and breathing hose

The strength test of a breathing hose mount part and a breathing hose shall be in accordance with either of **a)** and **b)** below.

- a) **Method applying constant force** The following two items shall be tested.
 - 1) Fix the respiratory inlet covering connected with a breathing hose on a rigid table or the like with a non-elastic belt, etc. At this time, take care so that the belt does not touch the breathing hose mount part. Then, apply a force shown in

Table 11 to the other end of breathing hose, and examine for any breakage or separation in the range from the breathing hose mount part to the other end of breathing hose.

- 2) Fix the cabinet containing the electric fan connected to the breathing hose on a rigid table or the like with a non-elastic belt, etc. At this time, take care so that the belt does not touch the breathing hose mount part. Then, apply a force shown in Table 11 to the other end of breathing hose, and examine for any breakage or separation in the range from the breathing hose mount part to the other end of breathing hose.

Table 11 Conditions of force

Shape	Tensile force N	Direction	Period of time s
Facepiece	100	Axial direction of breathing hose	10
Loose-fitting-type	50		

b) **Method using tensile tester** The following two items shall be tested.

- 1) Fix the respiratory inlet covering connected to the breathing hose to one of the fixing members of the tensile tester. Then, fix the other end of breathing hose to the other fixing member, pull them at a speed of 20 cm/min, and measure the force when the breathing hose shows any breakage or separation in the range from one breathing hose end to the other end. If the force applied by the tensile tester exceeds 100 N for the facepiece and 50 N for loose-fitting-type, and if no breakage or separation seems to occur easily with additional force application, the test may be interrupted by recording the final test force applied and the fact that no breakage or separation has occurred.
- 2) Fix the cabinet containing the electric fan connected to the breathing hose to one of the fixing members of the tensile tester. Then, fix the other end of breathing hose to the other fixing member, pull them at a speed of 20 cm/min, and measure the force when the breathing hose shows any breakage or separation in the range from one breathing hose end to the other end. If the force applied by the tensile tester exceeds 100 N for the facepiece and 50 N for loose-fitting-type, and if no breakage or separation seems to occur easily with additional force application, the test may be interrupted by recording the final test force applied and the fact that no breakage or separation has occurred.

9 Marking

9.1 PAPR

The following items shall be marked on PAPR at a conspicuous place.

- a) Product name or type of PAPR
- b) Number of this Standard
- c) Name of manufacturer or its general designation

- d) Year and month of manufacture or its abbreviation
- e) Grade of leakage rate
- f) Product name or symbol of particle filter

9.2 Particle filter

The following items shall be marked on a particle filter.

- a) Product name of particle filter
- b) Class according to filtering efficiency

9.3 Package for PAPR

The following items shall be marked on a package for PAPR.

- a) Classification Type of respiratory protective device/classification by shape/classification by performance of electric fan/grade of leakage rate/class of filtering efficiency of particle filter

Example 1 Respiratory protective device with electric fan/facepiece type, direct connection type or full facepiece/large air volume type/Grade S/PL3

Example 2 Respiratory protective device with electric fan/loose-fitting-type, front- or back-mounted type or hood/normal air volume type/Grade B/PS1

- b) Name or type of PAPR product
- c) Product name or symbol of particle filter (unique name or symbol different from other similar particle filters).

10 Instruction manual

10.1 PAPR

The instruction manual describing the following items shall be attached to PAPR.

- a) Product name or type of PAPR
- b) Number and title of this Standard
- c) Manufacturer's name
- d) Environmental condition under which PAPR can be used
- e) Environmental condition under which PAPR shall not be used
 - Example 1 Place where oxygen concentration possibly becomes under 18 %
 - Example 2 Place where toxic gas exists
- f) Statement for calling attention to the fact that PAPR loses of respiratory-protect function when PAPR is powered off (for loose-fitting-type PAPR)
- g) Type [describe **9.3 a)** and **9.3 c)**]
- h) Nominal service life (see Annex A)

- i) Minimum necessary air volume and maximum air volume (for loose-fitting-type PAPR)
- j) Mass
- k) Verifying method of air volume prior to use

Example 3 Verifying method that the air volume of electric fan or the like is not less than the air volume specified by the manufacturer using the exclusive airflow meter provided by the manufacturer

- l) Easy self check method of contact closeness between the wearer's face and the facepiece (for facepiece-type PAPR)
- m) Alert method of alarm
- n) Notes on charging when PAPR uses rechargeable batteries
- o) Notes on inspection, maintenance and storage
- p) Major components of PAPR Product name or symbol of the following major components (unique name or symbol different from other similar major components)
 - 1) Respiratory inlet covering
 - 2) Electric fan
 - 3) Particle filter
 - 4) Breathing hose (if used)
 - 5) Head harness for facepiece (if used)
 - 6) Rechargeable battery (if used)
- q) Notes on replacement of major components [e.g., "Use manufacturer-specified components"]
- r) Attaching/detaching method of particle filter
- s) Criteria of particle filter replacement
- t) Disposal method of PAPR (including influence on environment by materials)
- u) Basic idea for no hindrance to electric fan by water, dust and the like in the environment

10.2 Particle filter

The operation manual describing the following items shall be attached to particle filters.

- a) Notes on the effect that the instruction manual attached to PAPR should be read carefully.
- b) Replacement criteria
- c) Filter disposal method

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Annex A (informative)

Determination method of nominal service life

A.1 Determination method of nominal service life of facepiece-type PAPR

A.1.1 Specimen

Three pieces of facepiece-type PAPR shall be used. The power supply designated for PAPR shall be used. If the rechargeable batteries are to be used, they shall be fully charged. An unused product of particle filter designated for PAPR shall be used.

A.1.2 Test procedure

The test procedure shall be as follows.

- a) Measure the internal pressure of facepiece continuously on each specimen according to the method specified in **8.3**, and the period of time in which Equation (1) in **5.3** is continuously maintained (TM_1 , TM_2 and TM_3). In the procedure of **8.3.2 b**), if the specimen is provided with a manual flow control function, measure under the conditions at each of maximum and minimum flow rates within the range specific to the controller, and take the smaller value as the period of time of the specimen.
- b) Determine the arithmetic mean, TM_m , of TM_1 , TM_2 and TM_3 .
- c) The manufacturer shall designate the nominal service life taking TM_m as the upper limit.

A.2 Determination method of nominal service life of loose-fitting-type PAPR

A.2.1 Specimen

Three pieces of loose-fitting-type PAPR shall be used. The power supply designated for PAPR shall be used. If the rechargeable batteries are to be used, they shall be fully charged. An unused product of particle filter designated for PAPR shall be used.

A.2.2 Test procedure

The test procedure shall be as follows.

- a) Measure the air volume of PAPR continuously on each specimen according to the method specified in **8.4**, and the period of time in which the air volume not less than the relevant minimum necessary air volume in Table 4 of **5.4** is continuously maintained (TLF_1 , TLF_2 and TLF_3). In the procedure of **8.4.3 c**), if the specimen is provided with a manual flow control function, measure under the conditions at each of maximum and minimum flow rates within the range specific to the controller.

In this case, measure the air volume by adjusting the sucking flow rate of the sucking device so that the fine pressure differential gauge always maintains $0 \text{ Pa} \pm 5 \text{ Pa}$. In this case, an automatic adjusting device which always maintains the pressure at $0 \text{ Pa} \pm 5 \text{ Pa}$ should be used.

- b) Determine the arithmetic mean, TLF_m , of TLF_1 , TLF_2 and TLF_3 .
- c) The manufacturer shall designate the nominal service life taking TLF_m as the upper limit.

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